

WOULD YOU BUY A \$75,000 COFFIN?

CHE 396: Ethics Case

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CASE STUDY: AUTONOMOUS VEHICLES

An autonomous vehicle (AV), also known as a self-driving car, can safely drive with no human input using a programmed ability to sense the environment and its surroundings.

However, the programming in the AV can raise serious ethical concerns when the safety of the consumer and/or public is involved.

One of the major issues in AV programming is how the AV should respond when peoples' lives are at risk. More specifically, should the AV take action to save the driver or pedestrian(s)?

If YOU were the software engineer designing the AI for the autonomous vehicle, how would you program the safety software?

WHAT TYPES OF ISSUES DOES THIS PROBLEM RAISE?



This is an important issue because it involves the safety of the community, the liabilities of the autonomous vehicle producer, the privacy of the company, and is surrounded by important ethical questions.

This report includes the important ethical concerns and how they should be addressed when designing autonomous vehicles.

WHAT IF YOU WERE THE ENGINEER?

As a software engineer, you could be faced with making the tough decision of who would be saved in a catastrophic accident. Without a driver behind the wheel, the AV needs to “know” how to respond in an accident situation. For example, should the car turn and crash into a tree to avoid hitting a group of people?



[3]

FAST FORWARD 5 YEARS INTO THE FUTURE WHERE JAMES WILLIAM IS WORKING AT HIS DREAM COMPANY

As the lead software engineer for Deathla®, your manager Jasmine Marinara wants you to lead a team of engineers in programming accident avoidance algorithms into the AV. You bring up the issue of who should be put in danger if the only two options for avoiding an accident are putting pedestrians in danger or the passengers. She tells you that no one will buy an AV from us if it is programmed to kill them. She tells you that you must protect the passengers at all cost. You're deeply troubled as you think of the possibility of a group of children being in the way of the AV and getting injured due to your hand. You decide to weigh the possibilities to make the most ethical decision...



[3]

ETHICAL ISSUES: MORAL DECISION MAKING FROM THE AIChE CODE OF ETHICS

AIChE CODE OF ETHICS #1: Hold paramount the safety, health and welfare of the public and protect the environment in performance of their professional duties [2].



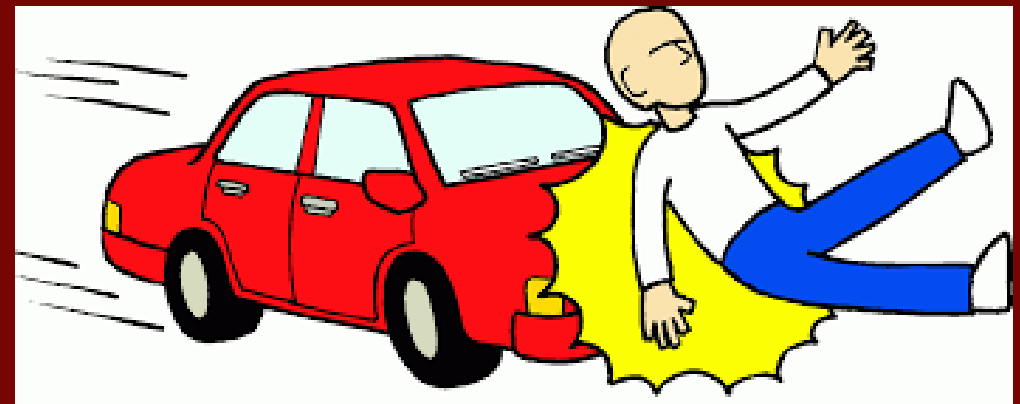
AIChE CODE OF ETHICS #2: Formally advise their employers or clients (and consider further disclosure, if warranted) if they perceive that a consequence of their duties will adversely affect the present or future health or safety of their colleagues or the public. [2].

1ST COURSE OF ACTION: SAFETY OF THE PASSENGERS

In compliance to your manager, you program the car to protect the safety of its passengers no matter what.

- ▶ The AV will calculate the velocity of the vehicle and other vehicles on the road as well as determine the spatial environment.
- ▶ If danger to the car arises, it will do a calculated risk assessment of possible plans of action and choose the one that will minimize the risk to the passengers.
- ▶ This program will treat objects (cars, humans, animals, etc.) with a hierarchy and will only choose to hit a person(s) if that is the **ONLY** way of protecting the passengers.

This course of action will come at the cost of knowing you programmed the car to possibly save 1 person over many.



[5]

2ND COURSE OF ACTION: SAFETY OF THE BYSTANDERS

Ruled by your conscience, you program the car to be altruistic; the car will save the most amount of people possible. The car will also be programmed to avoid hitting whoever is in the most in danger.

- ▶ The AV will assess the impact of hitting a pedestrian, who has no protection in an accident compared to the driver and will risk the life of the passengers to avoid hitting the pedestrian(s).
- ▶ If impact can be avoided with another vehicle (veering off road, into an inanimate object, etc), the AV will choose that, even if it harms the passenger.
- ▶ This program will put the safety of bystanders/pedestrians at a higher level than that of the passenger, allowing minimal harm to the public.

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DECISION: The public is at a higher risk when it comes to AV. Choosing to put the passenger's safety second can be minimized by implementing more internal safety features (air bags, survival kits, alerts to authorities, etc). By protecting those who may not be in an AV, code of ethics #1 is satisfied.

AICHE CODE OF ETHICS #2: Formally advise their employers or clients (and consider further disclosure, if warranted) if they perceive that a consequence of their duties will adversely affect the present or future health or safety of their colleagues or the public. [2].

DECISION: Companies will disclose all risks that come with a by-standard-first approach. Purchasers of the vehicles will be made aware of such risks, therefore satisfying code of ethics #2.

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THE TESLA MODEL 3 HAS FACED SCRUTINY

From a BBC article [6]:

"A preliminary report into a fatal accident involving a Tesla Model 3 in the US has found Autopilot had been engaged 10 seconds before the crash."

"According to the report, the driver did not appear to have his hands on the wheel and neither he nor the Autopilot took any evasive action."



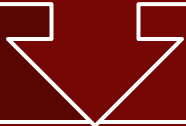
As you can see, the ethical programming in the AV needs to be considered because it can be involved in fatal accidents.

CONCLUDING THOUGHTS

Both companies and purchasers need to be aware of what risks come with developing technologies.



It is the decision of the company, overall, to produce the safest product out there.



As a part of this company, your decisions will impact lives daily. Choose wisely.

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